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	Human performance is easily impaired in a variety of aerospace tasks – such as remotely docking a spacecraft, or flying an aircraft in poor visibility, or finding and tracking a particular target airplane on a crowded air-traffic control display. These impairments occur because aerospace environments can degrade or remove the sensory cues used to drive and coordinate voluntary actions, and display systems may not be optimally matched to the human user. Fortunately, much progress has been made recently in the field of neuroscience toward understanding the physiological mechanisms responsible for voluntary motor behaviors. In particular, we are beginning to understand how the brain accomplishes visual search – the process that allows us to quickly respond to important items in our surroundings, even in the presence of other irrelevant stimuli. Understanding how individual neurons participate in this process is critical for identifying the factors that limit the speed and accuracy of visual search. Knowledge of these neural factors could be used to design visual displays that conform to the temporal and spatial constraints imposed by human neurobiology, thereby directly		

improving human performance in man-machine interfaces and virtual environments.
Arguably the most important class of movements made by primates during visual search are eye movements, because control of the position of the eyes determines which visual information is sampled and for how long. Decades of anatomical and physiological research have provided a detailed outline of the brain pathways that are responsible for the control of eye movements in primates. These pathways involve a network of areas including visual areas of the cerebral cortex (which provide information about the content of the visual scene), motor regions of the cerebellum and brainstem (which form the final motor commands), and several regions that lie in-between (and which possess a mixture of sensory and motor properties). It is these latter, intermediate, brain regions that are most likely responsible for forming the decision about where the eyes should move next during visual search, because they lie at the interface between the visual areas that provide evidence about the location of stimuli and the motor regions that execute the movements. One of these structures, the superior colliculus (SC), has long been known to play a pivotal role in the control of saccadic eye movements. However, most studies of the SC have focused on its contribution to the motor execution of saccades. Because of the importance of the SC in the decision processes that precede saccade generation, examination of this structure during visual search is likely to reveal fundamental aspects of the visual neural computations involved in identifying, localizing, and selecting visual targets. The main goal of this project would be to measure how neurons in the superior colliculus (SC) change their activity during a visual search task. Specifically, we propose to measure how the activity of these neurons is altered by the discriminability of visual targets and test whether these changes can predict the changes in the subjects' performance. Understanding how the information encoded by these neurons constrains overall search performance will foster the development of more predictive models of human performance and ultimately lead to the design of better aerospace displays and interfaces.

Task Description:

Goal: To develop a computational model of decision-making, involving multiple sources, agents and outputs, based on neurophysiological data.

The phenomenon: Visual search is a basic form of decision-making that is performed by humans and non-human primates several times a second (e.g., as you read this text). The brain pathways that accomplish this behavior have been identified, but the processing steps and computations that are involved in supporting this phenomenon are not known.

The experimental approach: By recording from single neurons in the brains of non-human primates as they perform visual search, we have a window into the internal processing steps that support visual search and how these change over time on a millisecond time scale – data that cannot be obtained from behavioral experiments with humans. In addition, we conduct parallel behavioral experiments with humans to confirm the applicability of this approach.

The hypothesis: By recording the activity of neurons during visual search, we can identify the biological computations that underlie performance in visual-motor tasks and decision-making.

Countermeasure readiness level: 2 at start of project, 3 now. The original hypothesis has now been validated and we are in the process of understanding in detail the neural processes steps that support neurally-mediated visual search. Specifically, we are testing how different sources of information are combined and read out by neurons during this basic decision task.

Expected countermeasure readiness level at end of project: 4 to 5. Once we understand the dynamic regulation of top-down and bottom-up factors, we will have identified the key features that are relevant for constructing biologically inspired models of visual search. We will then formulate how these features should be incorporated into computer-based models of decision-making (level 4), and then prove the feasibility and efficacy of this approach by constructing a model of visual search based on these biological features (level 5). After this initial demonstration, the concept may be applied to other processes that would benefit from automated decision-making (levels 5 – 6).

The results from these experiments provide new insights into the biological mechanisms underlying visual search behavior in humans and other primates. Although the immediate goals of the project were to expand our basic biological knowledge, the results are also be applicable to the stated goals of NASA's Space Human Factors, Intelligent Systems, and Aerospace Operation Systems Programs. Biological systems are capable of accomplishing tasks that have proved extremely difficult to duplicate using artificial intelligence and robotic systems – especially the job of detecting, discriminating and responding to stimuli quickly and accurately, even in the presence of background noise. Our results have provided new data about the algorithms and information processing streams used by biological systems to accomplish such tasks, and may therefore aid in the development of robust computational models of human visuomotor performance and in the design of the autonomous systems for future aerospace applications.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

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It has long been known that spatial cues have potent "attention" effects that are often attributed to increased sensitivity of sensory processing at the cued location. Alternatively, these effects could be caused by changing how information is weighted based on the cue, akin to an optimal observer, with no changes in the underlying sensitivity at the individual locations. Both models can explain the effects of cues on behavior, but they imply different neuronal mechanisms. We have now found that visuomotor neurons in the superior colliculus exhibit a neural correlate of the behavioral cueing effect. Moreover, this correlate is due to biases in activity caused by the presence or absence of the cue in the neurons' response field, with little change in the signal-to-noise ratio of processing at cued and uncued locations. These findings are consistent with the idea that the cueing effects we observed are caused by differential weighting of sensory evidence. A major goal of this project was to test which type of model best explains how prior information affects both the

Task Progress:	behavioral responses of subjects and the changes in activity of neurons at critical stages in the neural pathways. Our results indicate that the effects of prior information on the neurons we studied is not due to enhanced processing, but instead involves weighting evidence differently at the cued and uncued locations. These results also illustrate that the presence of a cueing effect does not, by itself, provide evidence that the sensitivity of sensory processing at the cued location has been increased. In addition, our results show that differential weighting not only provides a plausible alternative explanation for cueing effects, but also accounts for the attentional effects we found at the neuronal level. Although these results are somewhat technical, they are significant with respect to the CPR questions, because they provide a solid computational and biological foundation for developing models of human performance. The task we used involves combining incoming sensory inputs with prior information, a scenario that is very common in tasks that involve detecting, discriminating and responding to stimuli quickly and accurately. The ability of our model to account for both the behavioral and neuronal data indicates that the parameters of this model can provide biologically relevant measures of human performance.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
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